

---

# Information Retrieval

# Exercise

1. The following list of R's and N's represents relevant (R) and non-relevant (N) returned documents in a ranked list (the leftmost document is ranked the highest) of 20 documents retrieved in response to a query from a collection of 10,000 documents. Assume that there are 8 relevant documents in total in the collection.

R R N N N N N N R N R N N N R N N N N R

- a. Assume that these 20 documents are the system's complete result set. What is the MAP for the query?
- b. Assume that the system returned the entire 10,000 documents in a ranked list, and these are the first 20 results returned. What is the largest and smallest possible MAP that this system could have?

# Solution

---

a.  $\text{MAP} = 1/6 (1+1+3/9+4/11+5/15+6/20) = 0.555$

b.  $\text{MAP}_{\max} = 1/8 (1+1+3/9+4/11+5/15+6/20+7/21+8/22) = 0.503$

$\text{MAP}_{\min} = 1/8 (1+1+3/9+4/11+5/15+6/20+7/9999+8/10000) = 0.416$

# Information Retrieval Models

---

- **Boolean Model**

- ✓ term presence/absence → relevant/irrelevant, no ranking
- ✓ formalized query syntax using Boolean logic
- ✓ IR-systems of the **pre-web era**, as used in libraries, law enforcement etc.
- ✓ suitable for small collections

- **Vector Space Model (VSM)**

- ✓ queries and docs are represented in **a vector space**
- ✓ term weighting, scoring function → ranking
- ✓ free text queries
- ✓ web IR systems, used as add-on in classic environments

# Web Search vs. Standard IR

---

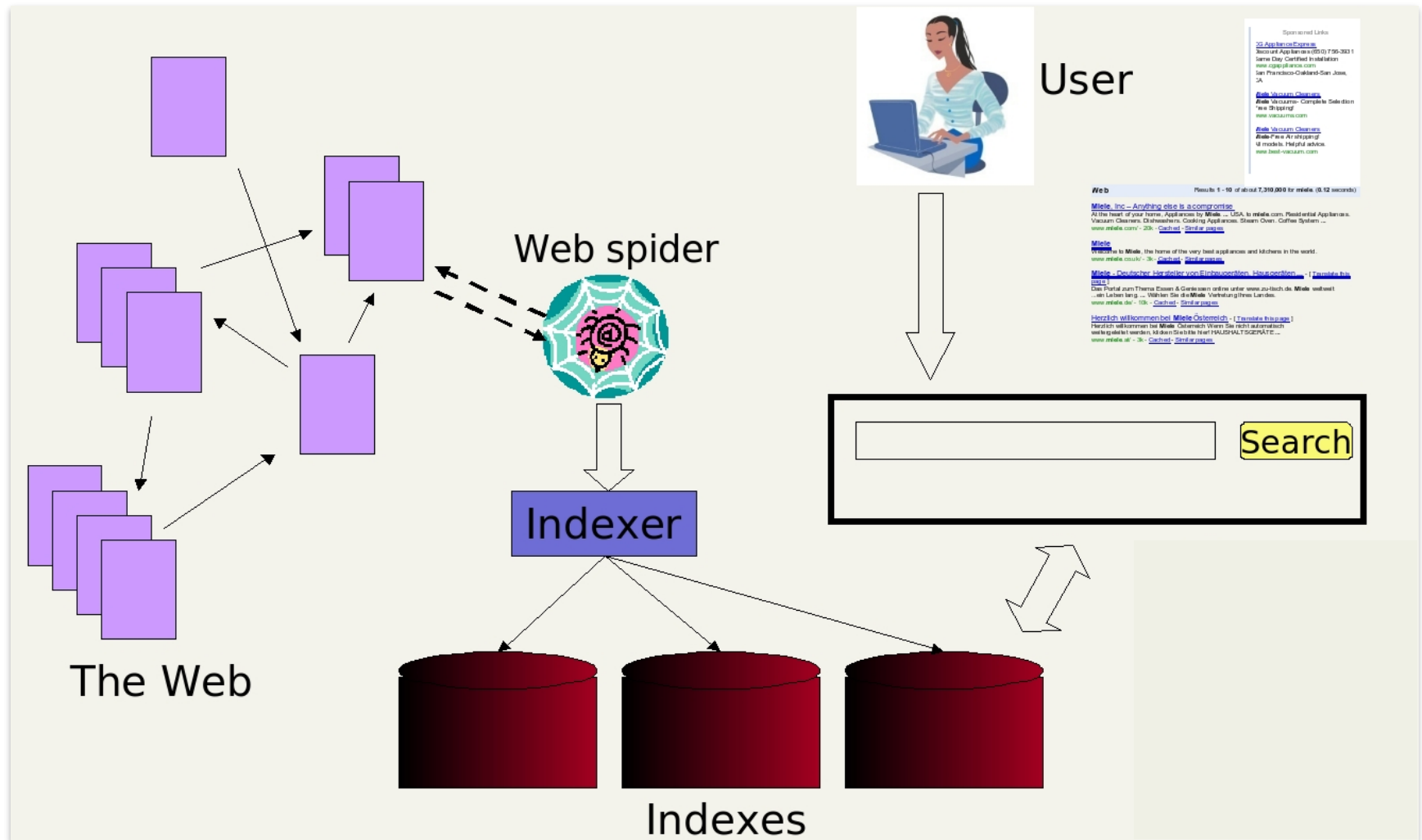
- Web Search (e.g. by Google) differs from standard IR:
  - Precision is more important than recall
  - On the web, recall cannot be measured anyway

# Web Search vs. Standard IR

---

- Web Search (e.g. by Google) differs from standard IR:
  - Precision is more important than recall
  - On the web, recall cannot be measured anyway
  - Web structure provides another basis for Webpage ranking
  - Web content may be outdated, fake, untrustworthy
  - The web is much bigger than any document collection, and changing very rapidly

# Standard Web Search Overview



# Web Search Phases

---

- Off-line
  - Crawling and indexing
- On-line
  - Querying and ranking



# Crawling

- Web Crawlers (aka spiders): traverse web graph and collect pages and their links
- Simplified web crawling algorithm:
  1. start with a processing queue containing the URL of 'seed pages'
  2. get the next URL from processing queue
  3. parse, extract outlinks (URLs), store page and link info
  4. mark current URL as processed
  5. put still unprocessed newly found outlinks into the processing queue
  6. run until all URLs are processed, and restart periodically
- Observe the change rate for pages to determine how often they should be re-crawled

# Indexing

---

- Non-trivial IR involves at least some linguistic analysis, e.g.
  - Stemming, Lemmatization, POS-Tagging, ...
- Many of these processing steps are computationally expensive.
- Due to size of the web and its heterogeneity, these cannot (yet) be applied to the entirety of documents

# Querying in Web Search

---

- Keyword-based search is the dominant paradigm
- Relevance
  - Vector space model and variants
- Query expansion
  - consists of selecting and adding terms to the user's query to minimize query-document mismatch and thereby improving retrieval performance.

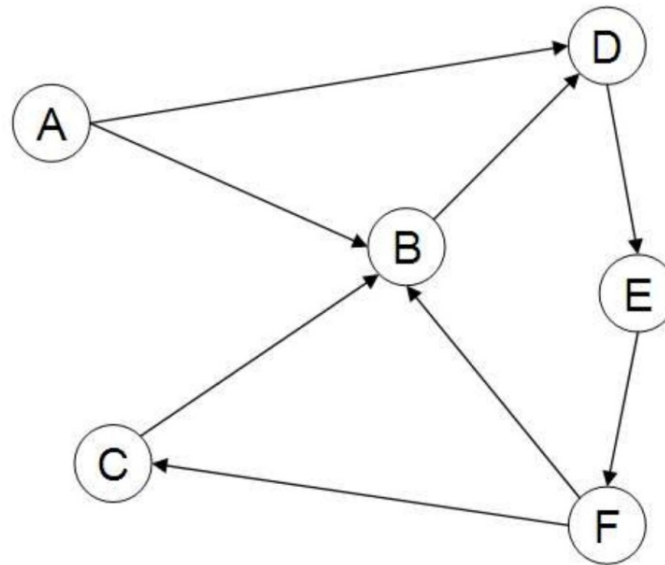
# Ranking in Web Search

---

- Web Pages are considered as documents
- Documents are not independent to each other
  - ✓ Connected via links
  - ✓ Represents a graph
- Ranks of the retrieved web pages do not only depend on textual similarity, there are other signals as well, like
  - ✓ Importance of that web page
    - Page Rank
    - Anchor text hyperlinked to that web page

# Ranking: PageRank

- Link structure provides information to measure 'importance' of a page
- Web = directed graph



- Inlinks: Pages that contain links to a page (in-degree)
  - ✓ On average between 8 to 15 (for entire web)
- Outlinks: Other pages that a page contains links to (out-degree)

# Ranking: PageRank

- Basis for [Google's PageRank algorithm](#) (Brin & Page 1998):
- Page rank  $PR_i$  of a page  $i$  increases with the number and page rank of the pages linking to it.

$$PR_i = \frac{1-d}{N} + d \sum_{\forall j \in \{(j,i)\}} \frac{PR_j}{C_j}$$

- $d$  = dampening factor,  $N$  = total number of pages,  $C_j$  = number of outlinks of page  $j$
- Recursive definition, Brin & Page (1998) describe iterative algorithm
  - About 100 iterations required for entire web

# Exercise

Compute the page ranks (damping factor = 0.8) for nodes in the graph in Figure. Initialize with a PageRank value of 0.2.

